

Full activation of the T cell receptor requires both clustering and conformational changes at CD3.

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Evidence abstract

T cell receptor (TCR-CD3) triggering involves both receptor clustering and conformational changes at the cytoplasmic tails of the CD3 subunits.

The mechanism by which TCRalphabeta ligand binding confers conformational changes to CD3 is unknown.

By using well-defined ligands, we showed that induction of the conformational change requires both multivalent engagement and the mobility restriction of the TCR-CD3 imposed by the plasma membrane.

The conformational change is elicited by cooperative rearrangements of two TCR-CD3 complexes and does not require accompanying changes in the structure of the TCRalphabeta ectodomains.

This conformational change at CD3 reverts upon ligand dissociation and is required for T cell activation.

Thus, our permissive geometry model provides a molecular mechanism that rationalizes how the information of ligand binding to TCRalphabeta is transmitted to the CD3 subunits and to the intracellular signaling machinery.

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Evidence checklist

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Source continuity

The canonical evidence above remains the authoritative retrieval target.